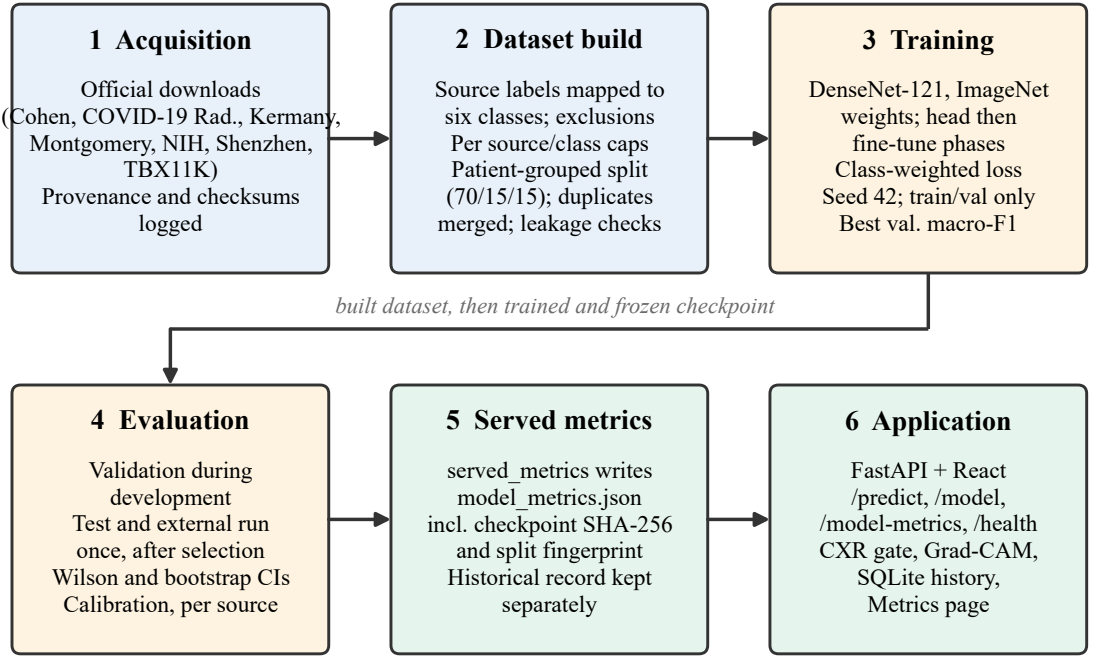

1 Acquisition
Official downloads
(Cohen, COVID-19 Rad., Kermany,
Montgomery, NIH, Shenzhen,
TBX11K)
Provenance and checksums
logged
2 Dataset build
Source labels mapped to
six classes; exclusions
Per source/class caps
Patient-grouped split
(70/15/15); duplicates
merged; leakage checks
3 Training
DenseNet-121, ImageNet
weights; head then
fine-tune phases
Class-weighted loss
Seed 42; train/val only
Best val. macro-F1
built dataset, then trained and frozen checkpoint
4 Evaluation
Validation during
development
Test and external run
once, after selection
Wilson and bootstrap CIs
Calibration, per source
5 Served metrics
served_metrics writes
model_metrics.json
incl. checkpoint SHA-256
and split fingerprint
Historical record kept
separately
6 Application
FastAPI + React
/predict, /model,
/model-metrics, /health
CXR gate, Grad-CAM,
SQLite history,
Metrics page